$$\begin{bmatrix}1&-2&-3\\-4&7&7\end{bmatrix}\xrightarrow[+4\times Row1]{Row2=Row2}\begin{bmatrix}1&-2&-3\\0&-1&-5\end{bmatrix}\xrightarrow[\times(-1)]{Row2=Row2}\begin{bmatrix}1&-2&-3\\0&1&5\end{bmatrix}$$

$$\xrightarrow[+2\times Row2]{Row1=Row1}\begin{bmatrix}1&0&7\\0&1&5\end{bmatrix}.\quad\therefore c_1=7 \text{ and } c_2=5.$$

By defn, the coordinate vector of x is $\begin{bmatrix}7\\5\end{bmatrix}$ □

5. $b_1=\begin{bmatrix}1\\5\\-3\end{bmatrix}, b_2=\begin{bmatrix}-3\\-7\\5\end{bmatrix}, x=\begin{bmatrix}4\\10\\-7\end{bmatrix}$

Solution: By defn. of Basis, $c_1b_1+c_2b_2\in H$ where $c_1,c_2\in R$.

Let, $x=c_1b_1+c_2b_2$. By Defn. of Ax, $x=\begin{bmatrix}b_1&b_2\end{bmatrix}\begin{bmatrix}c_1\\c_2\end{bmatrix}$.

By Theorem 3 from Chapter 1, the above matrix equation has the same solution set as the system of equations corresponding to the augmented matrix $\begin{bmatrix}b_1&b_2&x\end{bmatrix}$.

$$\begin{bmatrix}1&-3&4\\5&-7&10\\-3&5&-7\end{bmatrix}\xrightarrow[-5\times Row1]{Row2=Row2}\begin{bmatrix}1&-3&4\\0&8&-10\\-3&5&-7\end{bmatrix}\xrightarrow[+3\times Row1]{Row3=Row3}\begin{bmatrix}1&-3&4\\0&8&-10\\0&-4&5\end{bmatrix}$$

$$\xrightarrow[\times 1/2]{Row2=Row2}\begin{bmatrix}1&-3&4\\0&4&-5\\0&-4&5\end{bmatrix}\xrightarrow[+Row2]{Row3=Row3}\begin{bmatrix}1&-3&4\\0&4&-5\\0&0&0\end{bmatrix}$$

$\therefore c_1-3c_2=4$ and $4c_2=-5\Rightarrow c_2=-5/4$

$c_1+\frac{15}{4}=4\Rightarrow c_1=\frac{1}{4}$. By defn, the coordinate vector of x is

$\begin{bmatrix}1/4\\-5/4\end{bmatrix}$. □

Exercises 9–12 display a matrix A and an echelon form of A. Find bases for Col A and Nul A, and then state the dimensions of these subspaces.

9. $A=\begin{bmatrix}1&-3&2&-4\\-3&9&-1&5\\2&-6&4&-3\\-4&12&2&7\end{bmatrix}\sim\begin{bmatrix}\boxed{1}&-3&2&-4\\0&0&\boxed{5}&-7\\0&0&0&\boxed{5}\\0&0&0&0\end{bmatrix}$